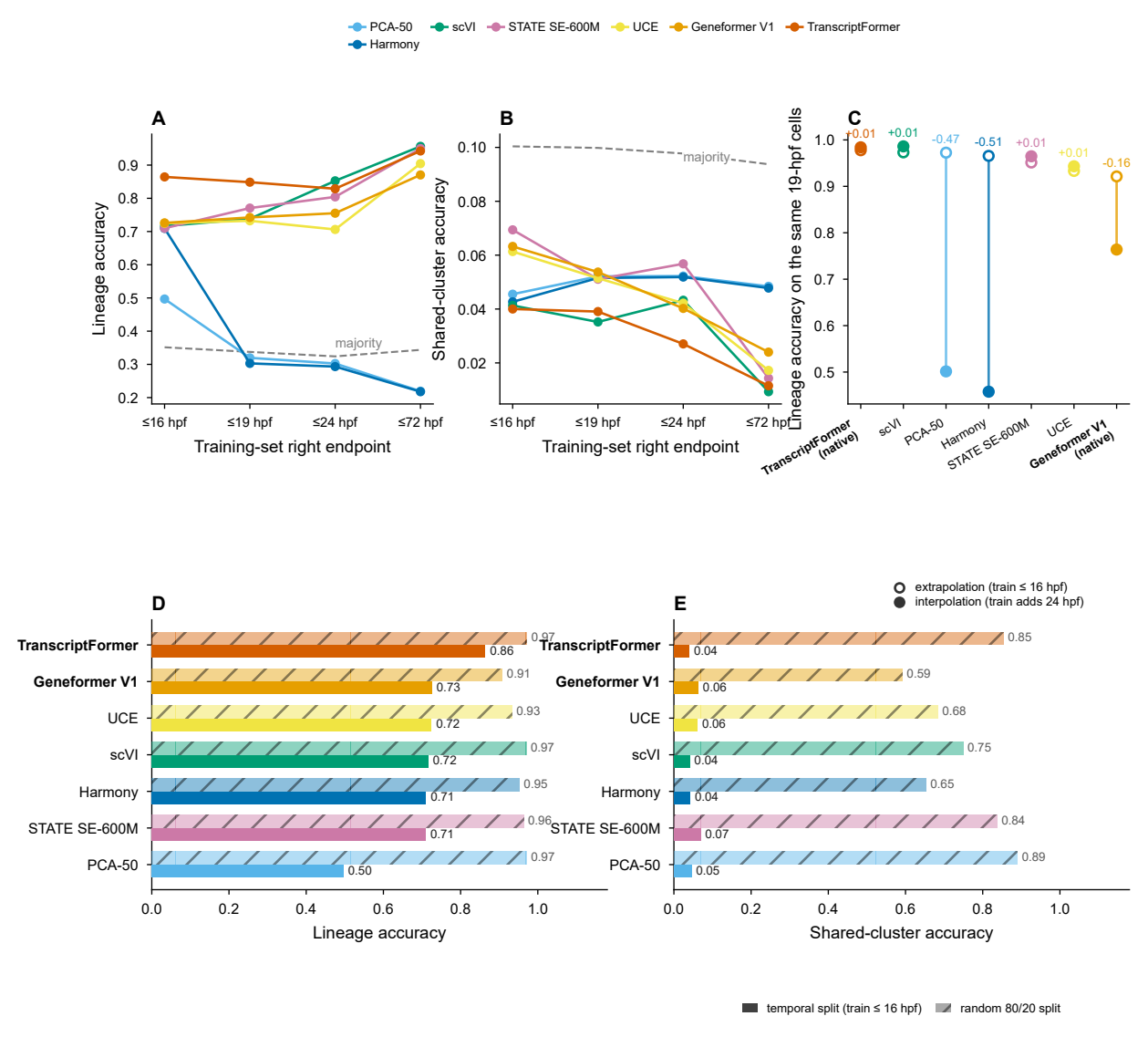

PCA-50
scVI
STATE SE-600M
UCE
Geneformer V1
TranscriptFormer
Harmony
A
Lineage accuracy
majority
≤16 hpf
≤19 hpf
≤24 hpf
≤72 hpf
Training-set right endpoint
B
Shared-cluster accuracy
majority
Training-set right endpoint
C
Lineage accuracy on the same 19-hpf cells
+0.01
+0.01
-0.47
-0.51
+0.01
+0.01
-0.16
TranscriptFormer (native)
scVI
PCA-50
Harmony
STATE SE-600M
UCE
Geneformer V1 (native)
D
TranscriptFormer 0.86 0.97
Geneformer V1 0.73 0.91
UCE 0.72 0.93
scVI 0.72 0.97
Harmony 0.71 0.95
STATE SE-600M 0.71 0.96
PCA-50 0.50 0.97
Lineage accuracy
E
extrapolation (train ≤ 16 hpf)
interpolation (train adds 24 hpf)
TranscriptFormer 0.04 0.85
Geneformer V1 0.06 0.59
UCE 0.06 0.68
scVI 0.04 0.75
Harmony 0.04 0.65
STATE SE-600M 0.07 0.84
PCA-50 0.05 0.89
Shared-cluster accuracy
temporal split (train ≤ 16 hpf)
random 80/20 split